Week 10 teacher guide

Scale, proportions, circles, and review

PowerPoint slides 869-966 | Student workbook pp. 102-112

A fixed multiplier connects scaled lengths and proportional quantities. Circle formulas use radius or diameter in different ways. The final weekdays revisit the entire program.

The seven days

Day 064 · pp. 102-104 · Saturday · 20 questions,including 2 challenges
Scale and proportional relationships

Day 065 · pp. 105-107 · Sunday · 20 questions,including 2 challenges
Circles and first previews

Day 066 · p. 108 · Monday · 20 questions,including 2 challenges
Scale, proportions, circles, and review

Day 067 · p. 109 · Tuesday · 20 questions,including 2 challenges
Scale, proportions, circles, and review

Day 068 · p. 110 · Wednesday · 25 questions,including 2 challenges
Scale, proportions, circles, and review

Day 069 · p. 111 · Thursday · 20 questions,including 2 challenges
Scale, proportions, circles, and review

Day 070 · p. 112 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

Each weekend lesson uses a three-page packet; each weekday uses one question page. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons and ordinary weekdays contain 20 questions including C1 and C2; Wednesday contains 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The weekend packet includes models and worked examples, or a scheduled assessment alternative. Read a worked example after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer review is optional after Friday and does not add to the 80-minute weekend lesson.

Built-in review

Day 066: Scale, proportion and circles (11, 12, 13, 14)

Day 067: Scale, proportion and circles (11, 12, 13, 14)

Day 068: Weekly puzzle (15)

Day 069: Data and statistics (11, 12, 13, 14)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Assessment alternatives: Final mixed review (p. 106). Replace ordinary practice or use a separate assessment sitting; do not add these on top of the assigned day.

Saturday. Scale and proportional relationships

Day 064 · pp. 102-104 | Slides 869-897

Goal: Use scale factors and identify relationships with a constant rate per one.

Check first: Ratios, unit rates, rectangle area, and coordinate graphs.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 869-870 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 871-872 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 873-874 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 875 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 876-878 · Explore · P4, P5, P6, P7, P8

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 879-881 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 882 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 883 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 884 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 885 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 886-887 · Guided practice · X1, X2

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 888-889 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 890-893 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 894 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 895 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 896 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 897 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: A constant step is not always a constant ratio

Watch and discuss; this is a projected investigation, not an extra assigned question.

Compare rules A: y = 3x and B: y = 3x + 4 for x = 0, 1, 2, 4. Build the tables and plot both lines. Both increase by 3 when x increases by 1. Decide which is proportional, and explain why equal steps alone are not enough.

1. Build both tables including the zero row.
2. Check ratios: A ratio is 3; B ratios vary.
3. Graph both lines; only A passes through the origin, matching proportionality.

Ask: What must be true at x = 0 in a proportional relationship?

If stuck: Compare the zero row first, then compute y/x for two nonzero rows.

Expected: A outputs 0, 3, 6, 12 and is proportional; k = 3. B outputs 4, 7, 10, 16 and is not proportional.

C1 · Challenge discussion

Ask: When a map is enlarged, which distance changes and which stays fixed?

If stuck: Write kilometers per centimeter for each map.

Expected: 27 km and 18 km; 45 km total. New scale 4 cm: 15 km (or 2 cm: 7.5 km).

C2 · Challenge discussion

Ask: How many dimensions are multiplied in an area calculation?

If stuck: Draw one scale arrow along each side.

Expected: 9 × 15 cm; perimeter 48 cm; area 135 square cm. Perimeter factor 1.5; area factor 2.25.

What to notice

Uses the same multiplier on every side. → Ask for a ratio for each matching side.

Squares the scale factor for area. → Count rows and columns of original copies.

Common error: Adds the same amount to every length. Repair: Compare each copy-to-original ratio.

Sunday. Circles and first previews

Day 065 · pp. 105-107 | Slides 898-928

Goal: Find circle measures and connect percent change and signed steps to earlier models.

Check first: Area, squares, percent of a whole, and negative-number order.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 898-899 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 900-901 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 902 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 903-905 · Explore · P4, P5, P6, P7, P8

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 906-908 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 909 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 910 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 911 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 912-917 · Guided practice · X1, X2, X3, X4, X5, X6

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 918-919 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 920-923 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 924 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 925 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 926 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 927 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Slides 928 · Assessment alternative

Replace ordinary practice with this check, or use a separate sitting. It is not extra assigned work.

Sunday. Discussion and support

Investigation T: Double a circle: edge versus interior

Watch and discuss; this is a projected investigation, not an extra assigned question.

Circle A has diameter 6 cm; circle B has diameter 12 cm. Predict the circumference and area factors before calculating. Find all four exact measures, and use 3.14 to check the areas. Explain why radius must be found before area.

1. Halve the diameters to identify radii 3 and 6.

2. Use $C = \pi d$: 6π and 12π, a factor of 2.

3. Use $A = \pi r^2$: 9π and 36π, a factor of 4; approximately 28.26 and 113.04.

Ask: Which measurement grows once, and which combines two scaled lengths?

If stuck: Label radii 3 and 6 before writing any area calculation.

Expected: Circle A: $C = 6\pi$ cm; area $= 9\pi \approx 28.26$ square cm. Circle B: $C = 12\pi$ cm; area $= 36\pi \approx 113.04$ square cm. Circumference factor 2; area factor 4.

C1 · Challenge discussion

Ask: Which formula counts surface, and which follows an edge?

If stuck: Find radius before circle area; keep the square area separate.

Expected: Deck $196 - 25\pi \approx 117.5$ square m; pool boundary $10\pi \approx 31.4$ m.

C2 · Challenge discussion

Ask: Which circle measurement determines distance in one turn?

If stuck: Draw one full turn as one circumference.

Expected: A: circumference 40π cm, area 400π square cm; B: circumference 80π cm, area 1600π square cm; B makes 6 turns.

What to notice

Marks the center and halves the diameter. → Ask which segment runs from center to circle.

Names linear versus square units. → Trace the edge, then shade the interior.

Common error: Squares the diameter. Repair: Mark two equal radii along the diameter.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Saturday

869. Week 10, Saturday
Watch and discuss

870. Today
Watch and discuss

871. Recall
Day 064 · Workbook p. 102 · R1, R2, R3, R4

872. Recall: answers
Day 064 · Workbook p. 102 · R1, R2, R3, R4

873. Whole-number scale factor
Day 064 · Workbook p. 102 · P1

874. Fractional scale factor
Day 064 · Workbook p. 102 · P2, P3

875. The model
Watch and discuss

876. Area multiplier is square of length multiplier
Day 064 · Workbook pp. 102, 103 · P4, P5

877. Graph through origin versus nonzero intercept
Day 064 · Workbook p. 103 · P6

878. Non-proportional story with fixed fee
Day 064 · Workbook p. 103 · P7, P8

879. Worked example · 1 of 3
Watch and discuss · worked example

880. Worked example · 2 of 3
Watch and discuss · worked example

881. Worked example · 3 of 3
Watch and discuss · worked example

882. Your turn
Day 064 · Workbook p. 103 · F

883. Two ways to see it
Watch and discuss · compare the same example

884. A claim to check
Day 064 · Workbook p. 103 · D

885. Words and meaning
Watch and discuss

886. Guided practice · X1
Day 064 · Workbook p. 103 · X1

887. Guided practice · X2
Day 064 · Workbook p. 103 · X2

888. A constant step is not always a constant ratio
Watch and discuss · draw on separate paper

889. A constant step is not always a constant ratio
Watch and discuss · draw on separate paper

890. Challenge C1
Day 064 · Workbook p. 104 · C1

891. Challenge C1 · Discuss
Day 064 · Workbook p. 104 · C1

892. Challenge C2
Day 064 · Workbook p. 104 · C2

893. Challenge C2 · Discuss
Day 064 · Workbook p. 104 · C2

894. In your own words
Day 064 · Workbook p. 104 · S

895. Exit ticket
Day 064 · Workbook p. 104 · E

896. The week ahead
Days 066-070 · Workbook pp. 108-112

897. Tutor reference
Tutor only;hidden in slide show

Sunday

898. Week 10, Sunday
Watch and discuss

899. Today
Watch and discuss

900. Circumference from diameter
Day 065 · Workbook p. 105 · P1

901. Circumference from radius
Day 065 · Workbook p. 105 · P2, P3

902. The model
Watch and discuss

903. Area from diameter
Day 065 · Workbook p. 105 · P4, P5

904. Percent increase and decrease
Day 065 · Workbook p. 105 · P6, P7

905. Positive addend
Day 065 · Workbook p. 105 · P8

906. Worked example · 1 of 3
Watch and discuss · worked example

907. Worked example · 2 of 3
Watch and discuss · worked example

908. Worked example · 3 of 3
Watch and discuss · worked example

909. Your turn
Day 065 · Workbook p. 105 · F

910. A claim to check
Day 065 · Workbook p. 105 · D

911. Words and meaning
Watch and discuss

912. Guided practice · X1
Day 065 · Workbook p. 105 · X1

913. Guided practice · X2
Day 065 · Workbook p. 105 · X2

914. Guided practice · X3
Day 065 · Workbook p. 105 · X3

915. Guided practice · X4
Day 065 · Workbook p. 105 · X4

916. Guided practice · X5
Day 065 · Workbook p. 105 · X5

917. Guided practice · X6
Day 065 · Workbook p. 105 · X6

918. Double a circle: edge versus interior
Watch and discuss · draw on separate paper

919. Double a circle: edge versus interior
Watch and discuss · draw on separate paper

920. Challenge C1
Day 065 · Workbook p. 107 · C1

921. Challenge C1 · Discuss
Day 065 · Workbook p. 107 · C1

922. Challenge C2
Day 065 · Workbook p. 107 · C2

923. Challenge C2 · Discuss
Day 065 · Workbook p. 107 · C2

924. In your own words
Day 065 · Workbook p. 105 · S

925. Exit ticket
Day 065 · Workbook p. 105 · E

926. The week ahead
Days 066-070 · Workbook pp. 108-112

927. Tutor reference
Tutor only;hidden in slide show

928. Optional assessment · Final mixed review
Day 065 · Workbook pp. 106-106 · Final mixed review

Answer review

929. Week 10 answers
This is the final answer section. Revisit any skill that needs more practice.

930. Scale and proportional relationships · 1 of 5
Day 064 · Workbook p. 102 · Items R1, R2, R3, R4, P1

931. Scale and proportional relationships · 2 of 5
Day 064 · Workbook pp. 102, 103 · Items P2, P3, P4, P5, P6

932. Scale and proportional relationships · 3 of 5
Day 064 · Workbook p. 103 · Items P7, P8, F, D, X1

933. Scale and proportional relationships · 4 of 5
Day 064 · Workbook pp. 103, 104 · Items X2, S, E

934. Scale and proportional relationships · 5 of 5
Day 064 · Workbook p. 104 · Items C1, C2

935. Circles and first previews · 1 of 5
Day 065 · Workbook p. 105 · Items P1, P2, P3, P4, P5

936. Circles and first previews · 2 of 5
Day 065 · Workbook p. 105 · Items P6, P7, P8, F, D

937. Circles and first previews · 3 of 5
Day 065 · Workbook p. 105 · Items X1, X2, X3, X4, X5

938. Circles and first previews · 4 of 5
Day 065 · Workbook pp. 105, 107 · Items X6, S, E, C1

939. Circles and first previews · 5 of 5
Day 065 · Workbook p. 107 · Items C2

940. Final mixed review · 1 of 3
Day 065 · Workbook p. 106 · Items 1, 2, 3, 4

941. Final mixed review · 2 of 3
Day 065 · Workbook p. 106 · Items 5, 6, 7, 8, 9

942. Final mixed review · 3 of 3
Day 065 · Workbook p. 106 · Items 10

943. Scale, proportions, circles, and review · 1 of 5
Day 066 · Workbook p. 108 · Items 1, 2, 3, 4, 5

944. Scale, proportions, circles, and review · 2 of 5
Day 066 · Workbook p. 108 · Items 6, 7, 8, 9, 10

945. Scale, proportions, circles, and review · 3 of 5
Day 066 · Workbook p. 108 · Items 11, 12, 13, 14, 15

946. Scale, proportions, circles, and review · 4 of 5
Day 066 · Workbook p. 108 · Items 16, 17, 18, C1

947. Scale, proportions, circles, and review · 5 of 5
Day 066 · Workbook p. 108 · Items C2

948. Scale, proportions, circles, and review · 1 of 4
Day 067 · Workbook p. 109 · Items 1, 2, 3, 4, 5

949. Scale, proportions, circles, and review · 2 of 4
Day 067 · Workbook p. 109 · Items 6, 7, 8, 9, 10

950. Scale, proportions, circles, and review · 3 of 4
Day 067 · Workbook p. 109 · Items 11, 12, 13, 14, 15

951. Scale, proportions, circles, and review · 4 of 4
Day 067 · Workbook p. 109 · Items 16, 17, 18, C1, C2

952. Scale, proportions, circles, and review · 1 of 6
Day 068 · Workbook p. 110 · Items 1, 2, 3, 4, 5

953. Scale, proportions, circles, and review · 2 of 6
Day 068 · Workbook p. 110 · Items 6, 7, 8, 9, 10

954. Scale, proportions, circles, and review · 3 of 6
Day 068 · Workbook p. 110 · Items 11, 12, 13, 14, 15

955. Scale, proportions, circles, and review · 4 of 6
Day 068 · Workbook p. 110 · Items 16, 17, 18, 19

956. Scale, proportions, circles, and review · 5 of 6
Day 068 · Workbook p. 110 · Items 20, 21, 22, 23, C1

957. Scale, proportions, circles, and review · 6 of 6
Day 068 · Workbook p. 110 · Items C2

958. Scale, proportions, circles, and review · 1 of 4
Day 069 · Workbook p. 111 · Items 1, 2, 3, 4, 5

959. Scale, proportions, circles, and review · 2 of 4
Day 069 · Workbook p. 111 · Items 6, 7, 8, 9, 10

960. Scale, proportions, circles, and review · 3 of 4
Day 069 · Workbook p. 111 · Items 11, 12, 13, 14, 15

961. Scale, proportions, circles, and review · 4 of 4
Day 069 · Workbook p. 111 · Items 16, 17, 18, C1, C2

962. Weekly check · 1 of 5
Day 070 · Workbook p. 112 · Items 1, 2, 3, 4, 5

963. Weekly check · 2 of 5
Day 070 · Workbook p. 112 · Items 6, 7, 8, 9, 10

964. Weekly check · 3 of 5
Day 070 · Workbook p. 112 · Items 11, 12, 13, 14

965. Weekly check · 4 of 5
Day 070 · Workbook p. 112 · Items 15, 16, 17, 18, C1

966. Weekly check · 5 of 5
Day 070 · Workbook p. 112 · Items C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 064 · Scale and proportional relationships

Workbook pp. 102-104 · assigned set

R1 · G-AREA

16 square cm

R2 · N-LCM

LCM 12

R3 · S-DISPLAY

2

Reason: Count every occurrence of 4.

R4 · G-SA

60 square cm

P1 · P-SCALE

8 cm by 12 cm; area 24 to 96 square cm, factor 4

Reason: Both dimensions double, so four originals fit in the copy.

P2 · P-SCALE

3, 4, 5 cm; corresponding angles stay equal

P3 · P-MAP

90 km

P4 · P-SCALE

45 square cm

P5 · P-PROP

k = 2.5; y = 2.5x

P6 · P-PROP

A: y = 2x. B is not proportional because it starts at 3 when x = 0.

Reason: Both have equal increases, but only A has zero output for zero input.

P7 · P-PROP

c = 3m + 4; not proportional

P8 · P-PROP

L = 1.5t, or L = $(\frac{3}{2})$ t

F · P-PROP

4; 3.5; 3.5

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · P-SCALE

The area is multiplied by 9 because both dimensions are tripled.

Reason: Applies one length multiplier to a two-dimensional measure. Tile a 3 by 3 array of original rectangles.

X1 · P-SCALE

6 × 4 cm; 24 square cm.

X2 · P-SCALE

$\frac{3}{2}$ square cm; area factor $\frac{1}{4}$.

Reason: Both dimensions are halved, so their product is multiplied by $\frac{1}{4}$.

S · EXPLAIN

Answers vary. A scaled copy multiplies every length by the same factor. Proportional relationships have the form y = kx.

Reason: Accept an accurate explanation supported by a matching example.

E · P-PROP

No. Ratios 5, $\frac{11}{3}$, $\frac{17}{5}$ differ. The rule y = 3x + 2 gives y = 2 at zero.

Reason: The change in y is always 6 when x increases by 2, but the ratio y to x is not constant.

C1 · P-MAP

27 km and 18 km; 45 km total. New scale 4 cm: 15 km (or 2 cm: 7.5 km).

Reason: The first scale is 7.5 km per cm. Multiply each map leg by 7.5. Doubling the drawing does not double the actual route; the represented kilometers per centimeter halve. Thus 4 new centimeters represent 15 km.

C2 · P-SCALE

9 × 15 cm; perimeter 48 cm; area 135 square cm. Perimeter factor 1.5; area factor 2.25.

Reason: Both dimensions multiply by 1.5, so area multiplies by 1.5 × 1.5 = 2.25. Original perimeter 32 becomes 48; original area 60 becomes 135. Multiplying area only once would give 90 and miss the second scaled dimension.

Day 065 · Circles and first previews

Workbook pp. 105-107 · assigned set

P1 · G-CIRCLE

8π cm; about 25.12 cm

P2 · G-CIRCLE

Diameter 6 cm; circumference 6π cm

P3 · G-CIRCLE

16π square cm; about 50.24 square cm

P4 · G-CIRCLE

9π square cm; about 28.26 square cm

P5 · G-CIRCLE

Both ratios equal π

Reason: Each circumference is π times its diameter. Dividing by the diameter leaves π.

P6 · P-CHANGE

25% increase; 20% decrease

P7 · P-CHANGE

75 tokens; the decrease uses the larger whole 100

Reason: First add 20. Then subtract 25 from 100. The percent applies to a new starting amount.

P8 · N-ADD

3; −7; −4

F · G-CIRCLE

8; 8; 64

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · G-CIRCLE

Radius = 7 cm, so area = 49π square cm.

Reason: Uses diameter where the formula requires radius. Mark the center and split the diameter into two equal radii.

X1 · G-CIRCLE

8π cm≈25.12 cm.

X2 · G-CIRCLE

9π square cm≈28.26 square cm.

X3 · G-CIRCLE

Radius 4 cm; area 16π square cm.

X4 · P-CHANGE

Increase 20; 25%.

X5 · P-CHANGE

$\frac{594}{5}$

X6 · N-ADD

Final −2; distance 12.

S · EXPLAIN

Answers vary. Circumference measures the boundary. Circle area uses the square of the radius. π is exact; 3.14 is an approximation.

Reason: Accept an accurate explanation supported by a matching example.

E · G-CIRCLE

Radius 6 cm; area 36π square cm; about 113.04 square cm

Reason: Half of 12 is 6. $A = \pi \times 6^2 = 36\pi$.

C1 · G-CIRCLE

Deck $196 - 25\pi \approx 117.5$ square m; pool boundary $10\pi \approx 31.4$ m.

Reason: The square area is 196. Pool radius 5 gives area 25π; subtract it from 196. Circumference uses π × diameter = 10π. Deck coverage uses square units; boundary length uses linear units.

C2 · G-CIRCLE

A: circumference 40π cm, area 400π square cm; B: circumference 80π cm, area 1600π square cm; B makes 6 turns.

Reason: Doubling radius doubles circumference but quadruples area. Wheel A travels $12 \times 40\pi = 480\pi$ cm. Each B turn covers 80π, so $480\pi \div 80\pi = 6$. The larger circle's area does not determine rolling distance.

Day 065 · Final mixed review

Workbook p. 106 · assessment alternative

1 · F-COMP

$\frac{5}{8} > \frac{3}{5}$

Reason: Rename as twenty-five fortieths and twenty-four fortieths.

2 · R-TABLE

21

3 · R-RATE

4 labels per minute; $\frac{1}{4}$ minute per label

4 · F-DIV

$\frac{5}{24}$ liter

5 · D-DIV

0.225

6 · E-DIST

2x + 14

7 · G-SA
60 square cm

8 · N-OPP
6

9 · S-IQR
Q1 2.5; Q3 9.5; IQR 7

10 · G-CIRCLE
3π cm; about 9.42 cm

Day 066 · Scale, proportions, circles, and review

Workbook p. 108 · assigned set

1 · N-FACT
36

2 · N-FACT
84

3 · N-FACT
48

4 · N-FACT
108

5 · F-SCALE
Larger; 8

Reason: Four thirds takes one whole six plus one third of six.

6 · R-RATE
0.65 dollars per marker

7 · D-SUB
8.714

8 · G-SA
84 square cm

9 · N-ORDER
−9, −4, 0

10 · S-MAD
2

11 · P-SCALE · review
4 cm and 6 cm

12 · P-MAP · review
80 km

13 · P-SCALE · review
5 square cm

14 · P-PROP · review
k = 1.5

15 · P-SCALE
The area is multiplied by 9 because both dimensions are tripled.

16 · P-SCALE
10 × 8 cm; 80 square cm.

17 · P-SCALE
5 square cm; area factor $\frac{1}{4}$.

Reason: Both dimensions are halved, so their product is multiplied by $\frac{1}{4}$.

18 · P-MAP
$\frac{45}{2}$ km.

C1 · P-MAP
Real 120 m; new model 30 cm; factor $\frac{5}{8}$.

Reason: First convert 48 × 2.5 = 120 m. At 4 m per cm, 120 ÷ 4 = 30 cm. Compare model lengths 30 ÷ 48 = $\frac{5}{8}$. A larger real-distance-per-centimeter scale creates a smaller model.

C2 · P-PROP
7.5 cups required; 1.5 cups short; $1.20.

Reason: The batch factor is 20 ÷ 8 = 2.5. Multiply 3 by 2.5 to get 7.5 cups. Subtract 6 to find 1.5 missing cups; 1.5 × 0.80 = $1.20.

Day 067 · Scale, proportions, circles, and review

Workbook p. 109 · assigned set

1 · F-EQUIV
0.5; 50%

2 · F-EQUIV
0.25; 25%

3 · F-EQUIV
0.75; 75%

4 · F-EQUIV
0.125; 12.5%

5 · F-LINE
At 1 $\frac{3}{4}$, the seventh quarter-step from 0

6 · R-COMP
Mix A

Reason: Use equivalent ratios with the same water amount, then compare juice.

7 · F-DIV
$4\frac{2}{3}$

8 · E-MULT

x = 3.2

9 · N-ABS

0 units from zero

Reason: Count the unit steps between the number and zero.

10 · S-MEDIAN

5

11 · G-CIRCLE · review

3π cm; about 9.42 cm

12 · G-CIRCLE · review

4π cm; about 12.56 cm

13 · G-CIRCLE · review

4π square cm; about 12.56 square cm

14 · G-CIRCLE · review

4π square cm; about 12.56 square cm

15 · G-CIRCLE

Radius = 7 cm, so area = 49π square cm.

16 · P-SCALE

12 × 10 cm; 120 square cm.

17 · P-SCALE

$\frac{15}{2}$ square cm; area factor $\frac{1}{4}$.

Reason: Both dimensions are halved, so their product is multiplied by $\frac{1}{4}$.

18 · P-MAP

33 km.

C1 · G-CIRCLE

Fence 41.96 m; cost $209.80.

Reason: Circumference = 2 × 3.14 × 7 = 43.96 m. Remove the 2 m opening to get 41.96. Multiply by $5 per meter. The gate replaces a boundary length, not a patch of ground.

C2 · P-CHANGE

Both $66.

Reason: A gives 80 × 0.75 × 1.10 = 66. B gives 80 × 1.10 × 0.75 = 66. Each step multiplies the current amount by a fixed factor; changing the multiplication order leaves the product unchanged.

Day 068 · Scale, proportions, circles, and review

Workbook p. 110 · assigned set

1 · N-SCALE

122

2 · N-SCALE

12

3 · N-SCALE

25

4 · N-SCALE

36

5 · F-COMP

$\frac{2}{4} > \frac{2}{5}$

Reason: With the same numerator, the fraction with larger pieces is larger.

6 · R-WHOLE

10 yellow; 25 purple

7 · P-PART

42 points

8 · F-DIV

$\frac{2}{5}$

Reason: Two and one half is five halves. Five halves times two fifths equals one.

9 · D-ADD

6.528

10 · E-MULT

x = 3.2

11 · G-SA

58 square cm

12 · E-INEQ

x < 5; no

13 · S-SPREAD

Both means 4. Range A 2; range B 6. B is more spread out.

Reason: Find the center, then compare the endpoint gaps.

14 · G-CIRCLE

3π cm; about 9.42 cm

15 · P-SCALE · review

New drawing: 1.5 cm by 2 cm. Actual area: 1,200 square m.

16 · P-SCALE

14 × 4 cm; 56 square cm.

17 · P-SCALE

$\frac{7}{2}$ square cm; area factor $\frac{1}{4}$.

Reason: Both dimensions are halved, so their product is multiplied by $\frac{1}{4}$.

18 · P-MAP
$\frac{35}{2}$ km.

19 · P-PROP
k = 7; y = 35.

20 · P-PROP
c = 7 + 2h; not proportional.

21 · P-PROP
L = $\frac{7}{2}$ t; 14 liters.

22 · G-CIRCLE
14π cm≈43.96 cm.

23 · G-CIRCLE
4π square cm≈12.56 square cm.

C1 · G-CIRCLE
439.6 m; 62.8 m/min.

Reason: One lap is 40 × 3.14 = 125.6 m. Three and a half laps give 125.6 × 3.5 = 439.6. Divide by 7 minutes to get 62.8 m/min. Use circumference for each lap, not circle area.

C2 · S-MEAN
Original mean $25; new order $35; new median $26.

Reason: Original total 100 gives mean 25. Five orders at mean 27 require total 135, so the new order costs 35. Ordered prices 18, 22, 26, 34, 35 have middle 26. The larger prices pull the mean above the median.

Day 069 · Scale, proportions, circles, and review

Workbook p. 111 · assigned set

1 · N-EST
1200

2 · N-EST
13

3 · N-EST
About 100

4 · N-EST
About 20

5 · R-EQUIV
No

6 · P-PART
42 points

7 · F-DIV
$2\frac{1}{2}$; stories vary

8 · D-DIV
1.2

9 · E-ADD
x = 17

10 · G-AREA
21 square cm

11 · S-MEAN · review
4

12 · S-MEDIAN · review
6

13 · S-RANGE · review
10

14 · S-MAD · review
$\frac{3}{2}$

15 · P-MAP
100 km

16 · G-CIRCLE
Radius = 7 cm, so area = 49π square cm.

17 · N-SIGNED
−12; 4

18 · R-PERCENT
25%

C1 · N-ADD
Final −1; total travel 14 floors; distance from zero 1.

Reason: Move from −3 to 5, then from 5 to −1. The route lengths add 8 + 6 = 14. Final position describes location; absolute value 1 measures only the final gap to zero, not the full route.

C2 · P-SCALE
Height 7.5 cm; area 45 square cm.

Reason: Length factor = 12 ÷ 8 = 1.5, so height 5 × 1.5 = 7.5. Area 12 × 7.5 ÷ 2 = 45. Both base and height scale, so area 20 × 1.5^2 = 45; the student multiplied only one dimension.

Day 070 · Weekly check

Workbook p. 112 · assigned set

1 · F-MULT
$\frac{3}{5}$

2 · R-GRAPH
Yes

Reason: Every vertical value is twice the horizontal value.

3 · P-PART

8 stickers

4 · F-DIV

1 meters

5 · D-MULT

0.0100

6 · E-DIST

3x + 18

7 · G-SA

58 square cm

8 · N-OPP

1

9 · S-MAD

3

10 · P-SCALE

The area is multiplied by 9 because both dimensions are tripled.

11 · P-SCALE

18 × 8 cm; 144 square cm.

12 · P-SCALE

9 square cm; area factor $\frac{1}{4}$.

Reason: Both dimensions are halved, so their product is multiplied by $\frac{1}{4}$.

13 · P-MAP

$\frac{81}{2}$ km.

14 · P-PROP

k = 9; y = 63.

15 · P-PROP

c = 9 + 4h; not proportional.

16 · P-PROP

L = $\frac{9}{4}$ t; 18 liters.

17 · G-CIRCLE

18π cm≈56.52 cm.

18 · G-CIRCLE

16π square cm≈50.24 square cm.

C1 · G-CIRCLE

Paving 83.44 square m; discounted cost $750.96.

Reason: Rectangle area 96; fountain radius 2 gives area 3.14 × 4 = 12.56. Paving area 96 − 12.56 = 83.44. Full cost 834.40; pay 90% = 750.96. Apply the discount to the bill after measuring only the paved region.

C2 · P-PROP

20 pages: A $10, B $7; 60 pages: A $18, B $21; agree at 40 pages, $14. Only B is proportional.

Reason: A has a $6 fixed fee but is $0.15 cheaper per page. It takes 6 ÷ 0.15 = 40 pages to close the fee gap. B has constant cost per page and costs 0 at 0 pages; A starts at $6, so it is not proportional.